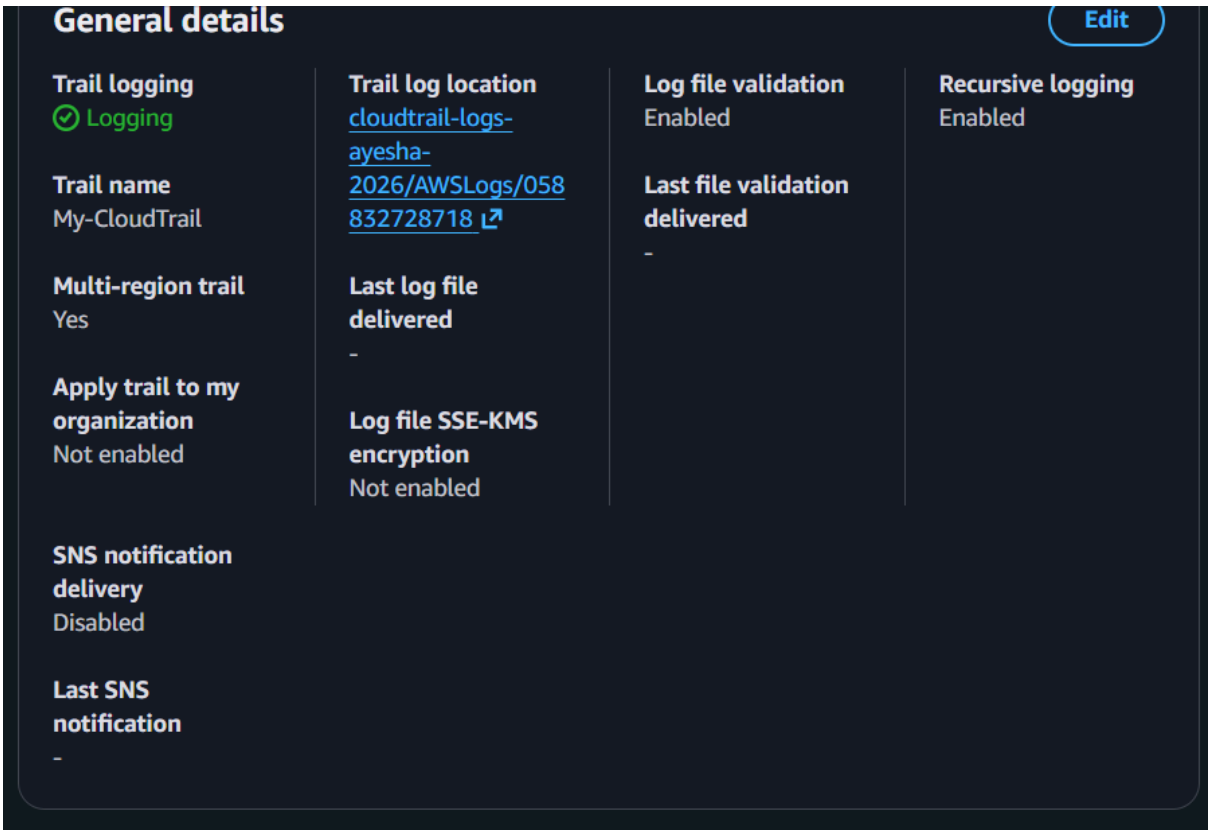
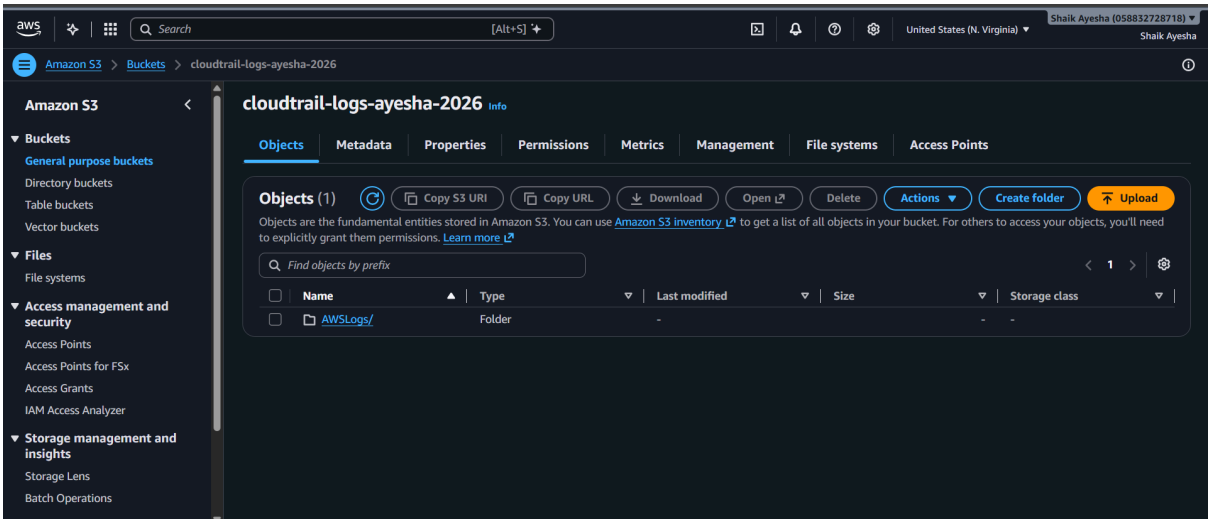
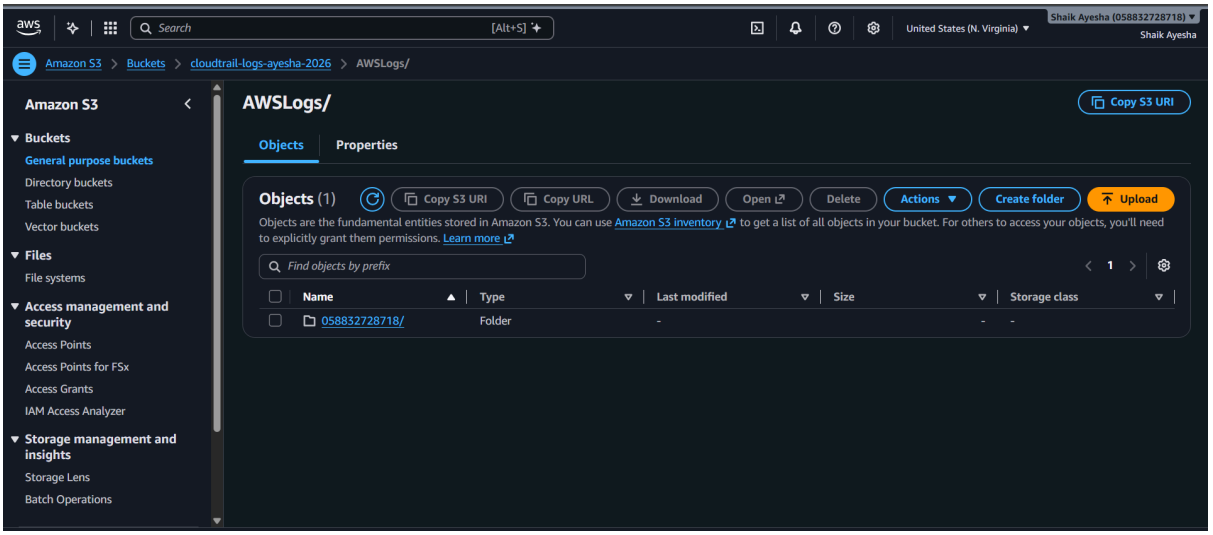
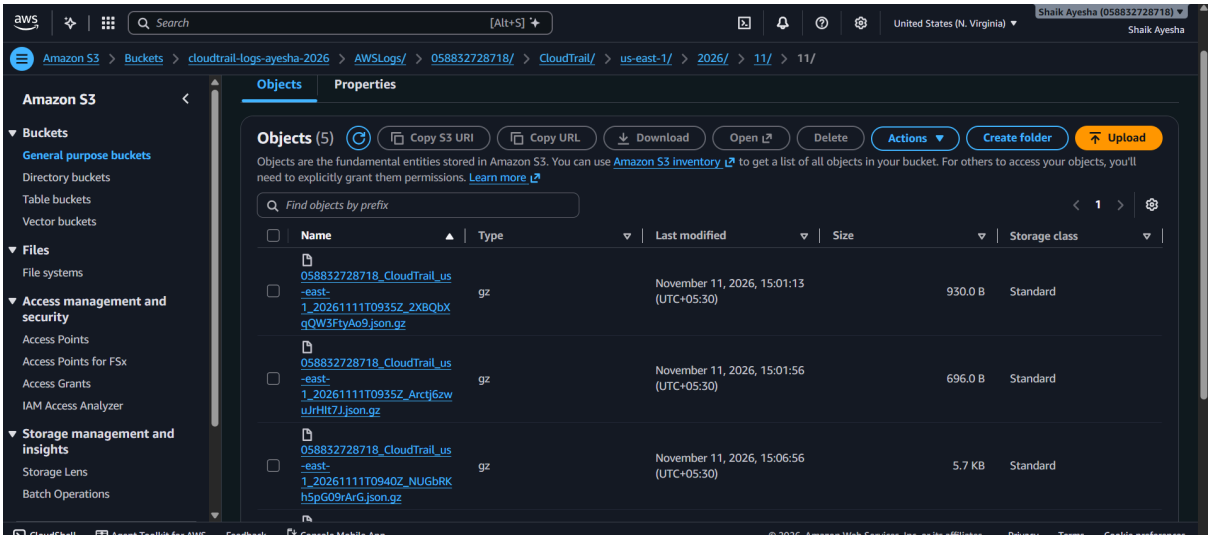
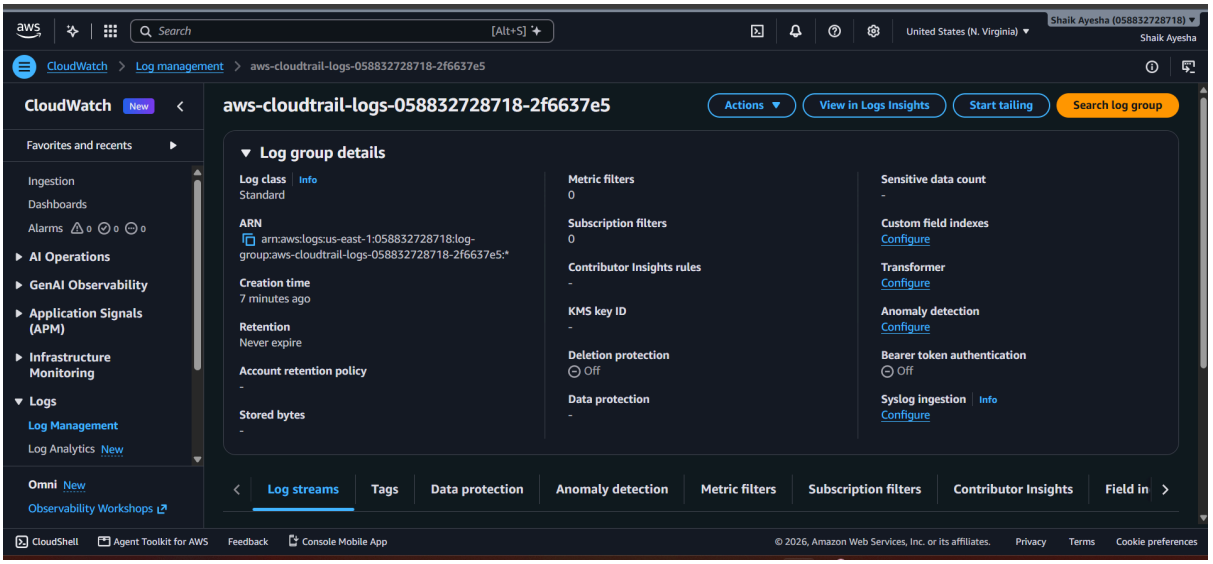
1. **Enable CloudTrail monitoring and store the events in S3 and CloudWatch log events**

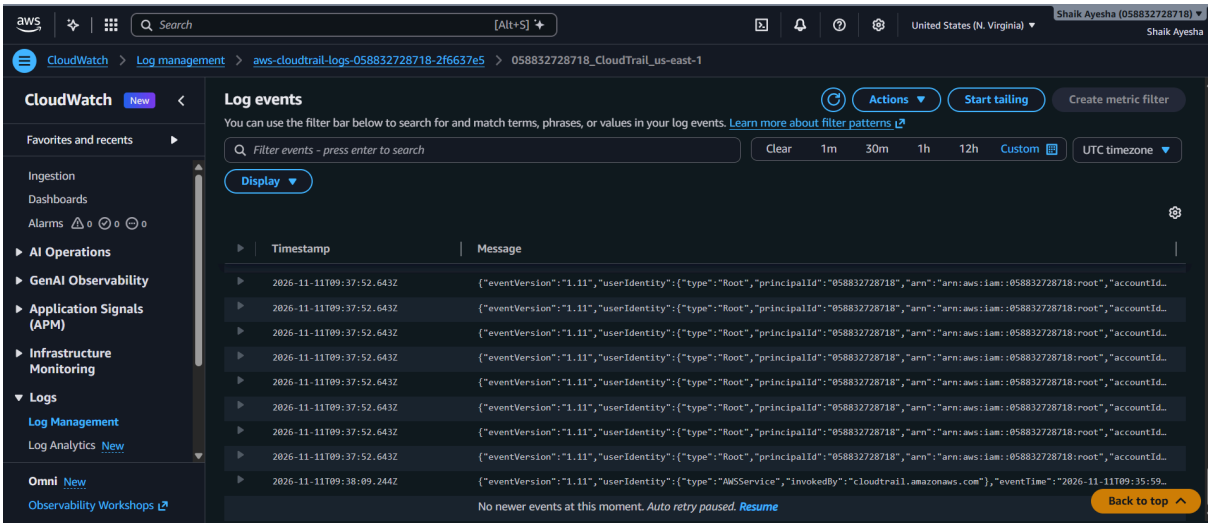





Search [Alt+S]
United States (N. Virginia)
Shaik Ayesha (058832728718)
Shaik Ayesha
Amazon S3 > Buckets > cloudtrail-logs-ayesha-2026 > AWSLogs/
Amazon S3
Buckets
General purpose buckets
Directory buckets
Table buckets
Vector buckets
Files
File systems
Access management and security
Access Points
Access Points for FSx
Access Grants
IAM Access Analyzer
Storage management and insights
Storage Lens
Batch Operations
AWSLogs/
Copy S3 URI
Objects
Properties
Objects (1)
Copy S3 URI
Copy URL
Download
Open
Delete
Actions
Create folder
Upload
Objects are the fundamental entities stored in Amazon S3. You can use Amazon S3 inventory to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. Learn more
Find objects by prefix
Name
Type
Last modified
Size
Storage class
058832728718/
Folder


Search [Alt+S]
United States (N. Virginia)
Shaik Ayesha (058832728718)
Shaik Ayesha
Amazon S3 > Buckets > cloudtrail-logs-ayesha-2026 > AWSLogs/ > 058832728718/ > CloudTrail/ > us-east-1/ > 2026/ > 11/ > 11/
Amazon S3
Objects
Properties
Objects (5)
Copy S3 URI
Copy URL
Download
Open
Delete
Actions
Create folder
Upload
Objects are the fundamental entities stored in Amazon S3. You can use Amazon S3 inventory to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. Learn more
Find objects by prefix
Name
Type
Last modified
Size
Storage class
058832728718_CloudTrail_us-east-1_20261111T0935Z_2XBQbXqQW3FtyAo9.json.gz
gz
November 11, 2026, 15:01:13 (UTC+05:30)
930.0 B
Standard
058832728718_CloudTrail_us-east-1_20261111T0935Z_Arctj6zwuJrHlt7J.json.gz
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November 11, 2026, 15:01:56 (UTC+05:30)
696.0 B
Standard
058832728718_CloudTrail_us-east-1_20261111T0940Z_NUGbRKh5pG09rArG.json.gz
gz
November 11, 2026, 15:06:56 (UTC+05:30)
5.7 KB
Standard


Search [Alt+S]
United States (N. Virginia)
Shaik Ayesha (058832728718)
Shaik Ayesha
CloudWatch > Log management > aws-cloudtrail-logs-058832728718-2f6637e5
CloudWatch New
Favorites and recents
Ingestion
Dashboards
Alarms
AI Operations
GenAI Observability
Application Signals (APM)
Infrastructure Monitoring
Logs
Log Management
Log Analytics New
Omni New
Observability Workshops
aws-cloudtrail-logs-058832728718-2f6637e5
Actions
View in Logs Insights
Start tailing
Search log group
Log group details
Log class Info
Standard
ARN
arn:aws:logs:us-east-1:058832728718:log-group:aws-cloudtrail-logs-058832728718-2f6637e5:*
Creation time
7 minutes ago
Retention
Never expire
Account retention policy
-
Stored bytes
-
Metric filters
0
Subscription filters
0
Contributor Insights rules
-
KMS key ID
-
Deletion protection
Off
Data protection
-
Sensitive data count
-
Custom field indexes
Configure
Transformer
Configure
Anomaly detection
Configure
Bearer token authentication
Off
Syslog ingestion Info
Configure
Log streams
Tags
Data protection
Anomaly detection
Metric filters
Subscription filters
Contributor Insights
Field in
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences



Step 1: Create an S3 Bucket

1. I logged in to the AWS Management Console.
2. I searched for Amazon S3.
3. I clicked on Create bucket.
4. I entered the bucket name cloudtrail-logs-ayesha-2026.
5. I selected the AWS region (US East – N. Virginia).
6. I kept Block all public access enabled.
7. I kept ACLs disabled.
8. I clicked Create bucket.

Step 2: Create a CloudTrail Trail

1. I searched for CloudTrail in the AWS Console.
2. I selected Trails from the left navigation panel.
3. I clicked Create trail.
4. I entered the trail name as My-CloudTrail.
5. I enabled Apply trail to all regions to monitor activities across all AWS regions.
6. I selected the existing S3 bucket cloudtrail-logs-ayesha-2026 as the storage location.
7. I kept log file SSE-KMS encryption disabled.
8. I enabled Log file validation to verify the integrity of delivered log files.
9. I kept Recursive logging enabled.
10. I left SNS notification delivery disabled.
11. I clicked Next.

Step 3: Configure Event Types

1. I selected Management events.
2. I enabled both Read and Write management events.
3. I kept the default event selection settings.

4. I reviewed the configuration.
5. I clicked Create trail.

Step 4: Verify CloudTrail Trail Creation

1. I navigated to CloudTrail → Trails.
2. I selected My-CloudTrail.
3. I verified that Trail logging was enabled.
4. I confirmed that Multi-region trail was set to Yes.
5. I verified that my S3 bucket was configured as the log storage location.
6. I confirmed that Log file validation was enabled.

Step 5: Configure CloudWatch Logs

1. I opened my created CloudTrail trail.
2. I navigated to the CloudWatch Logs section.
3. I clicked Edit.
4. I enabled CloudWatch Logs integration.
5. I selected the log group:
   aws-cloudtrail-logs-058832728718-2f6637e5
6. I selected the IAM role:
   CloudTrail_CloudWatch_Role
7. I saved the configuration.

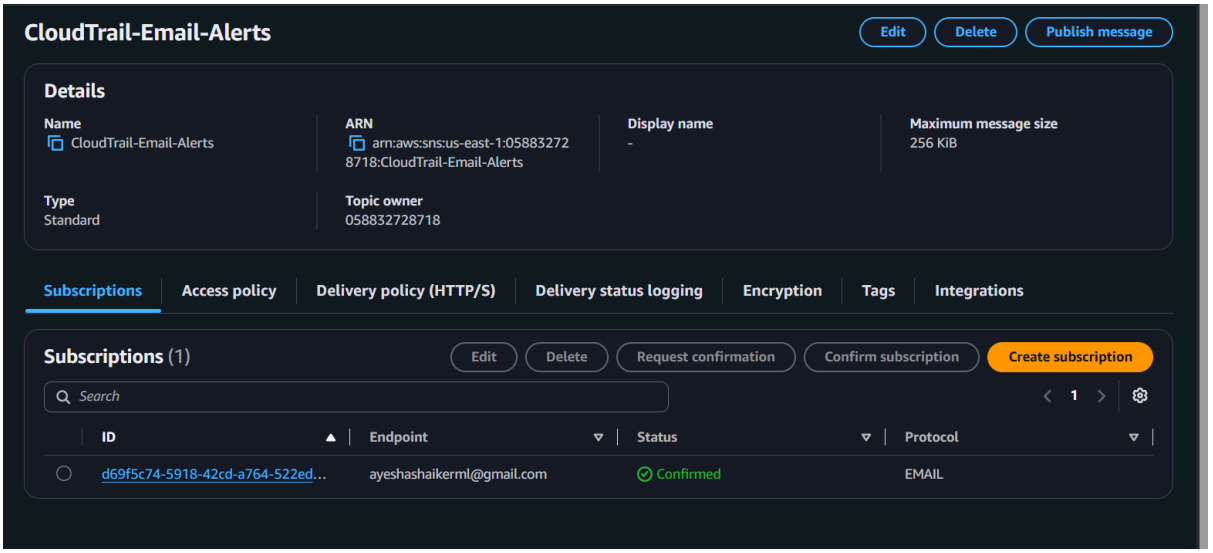
Step 6: Verify CloudWatch Logs

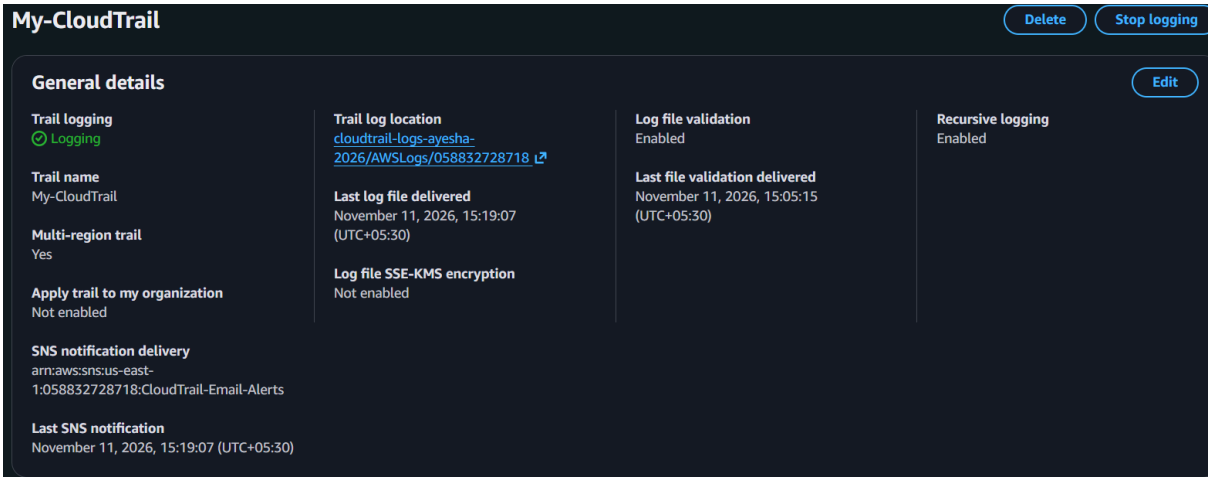
1. I opened Amazon CloudWatch.
2. I navigated to Logs → Log groups.
3. I searched for my CloudTrail log group.
4. I opened the log group.
5. I checked the available log streams.
6. I verified the CloudTrail events in the log streams.

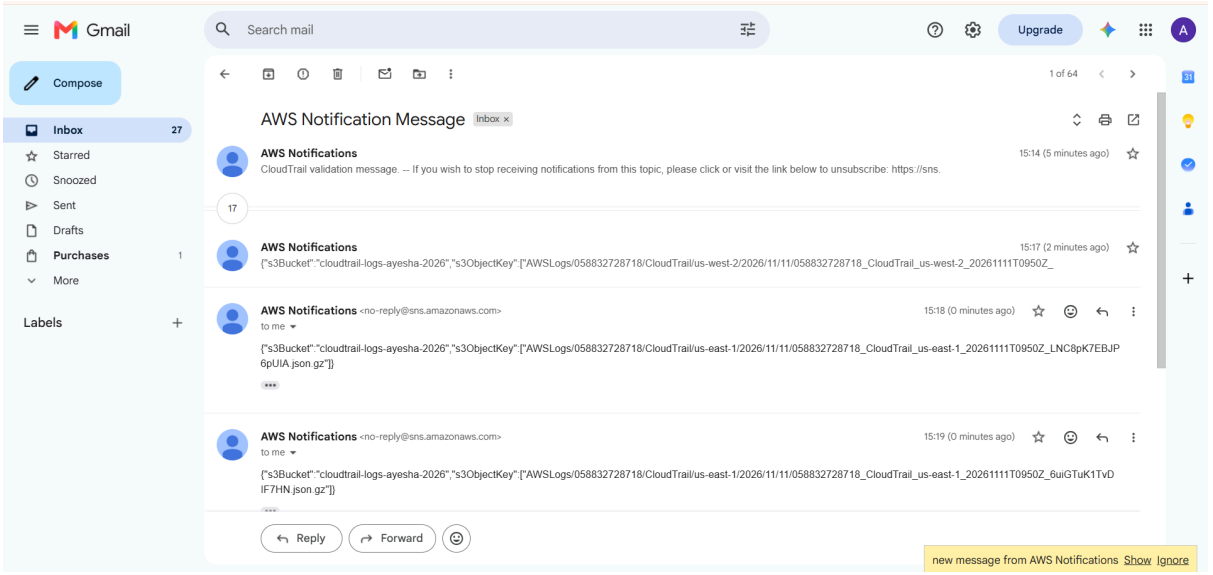
Step 7: Verify CloudTrail Logs in S3

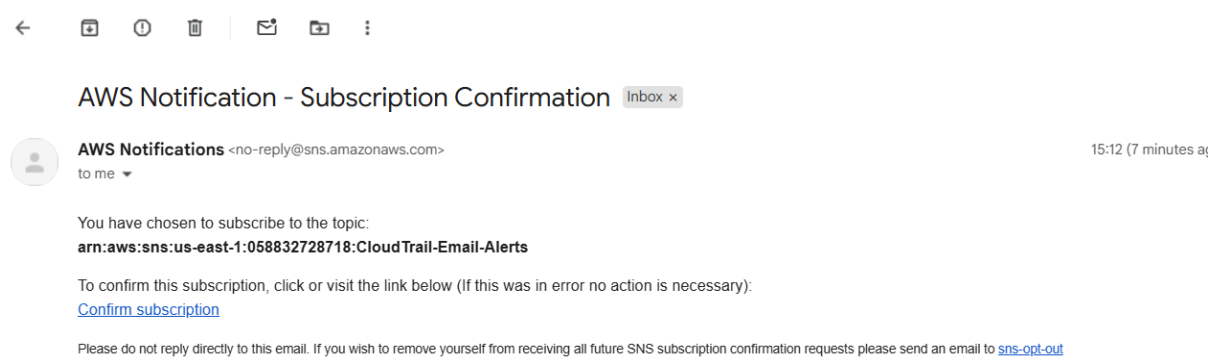
1. I opened Amazon S3.
2. I selected my bucket cloudtrail-logs-ayesha-2026.
3. I opened the AWSLogs folder.
4. I navigated to my AWS account ID folder.
5. I opened the CloudTrail folder.
6. I checked for the delivered CloudTrail log files.

2. Enable SNS for CloudTrail to send alerts via email


CloudTrail-Email-Alerts
Edit
Delete
Publish message
Details
Name
CloudTrail-Email-Alerts
ARN
arn:aws:sns:us-east-1:058832728718:CloudTrail-Email-Alerts
Display name
-
Maximum message size
256 KiB
Type
Standard
Topic owner
058832728718
Subscriptions
Access policy
Delivery policy (HTTP/S)
Delivery status logging
Encryption
Tags
Integrations
Subscriptions (1)
Edit
Delete
Request confirmation
Confirm subscription
Create subscription
Search
1
ID
Endpoint
Status
Protocol
d69f5c74-5918-42cd-a764-522ed...
ayeshashaikerml@gmail.com
Confirmed
EMAIL


My-CloudTrail
Delete
Stop logging
General details
Edit
Trail logging
Logging
Trail name
My-CloudTrail
Multi-region trail
Yes
Apply trail to my organization
Not enabled
SNS notification delivery
arn:aws:sns:us-east-1:058832728718:CloudTrail-Email-Alerts
Last SNS notification
November 11, 2026, 15:19:07 (UTC+05:30)
Trail log location
cloudtrail-logs-ayesha-2026/AWSLogs/058832728718
Last log file delivered
November 11, 2026, 15:19:07 (UTC+05:30)
Log file SSE-KMS encryption
Not enabled
Log file validation
Enabled
Last file validation delivered
November 11, 2026, 15:05:15 (UTC+05:30)
Recursive logging
Enabled





Step 1: Create an SNS Topic

1. I opened Amazon SNS.
2. I selected Topics.
3. I clicked Create topic.
4. I selected Standard.
5. I entered the topic name CloudTrail-Alerts.
6. I clicked Create topic.

Step 2: Create Email Subscription

1. I opened the CloudTrail-Alerts topic.
2. I clicked Create subscription.
3. I selected Email as the protocol.
4. I entered my email address.
5. I clicked Create subscription.

Step 3: Confirm Email Subscription

1. I opened my email inbox.
2. I opened the AWS SNS confirmation email.
3. I clicked Confirm subscription.
4. I verified that the subscription status changed to Confirmed.

Step 4: Enable SNS in CloudTrail

1. I opened CloudTrail → Trails.
2. I selected My-CloudTrail.
3. I clicked Edit under General details.
4. I scrolled to Additional settings.
5. I enabled SNS notification delivery.
6. I selected Existing SNS topic.
7. I selected CloudTrail-Alerts.
8. I saved the changes.

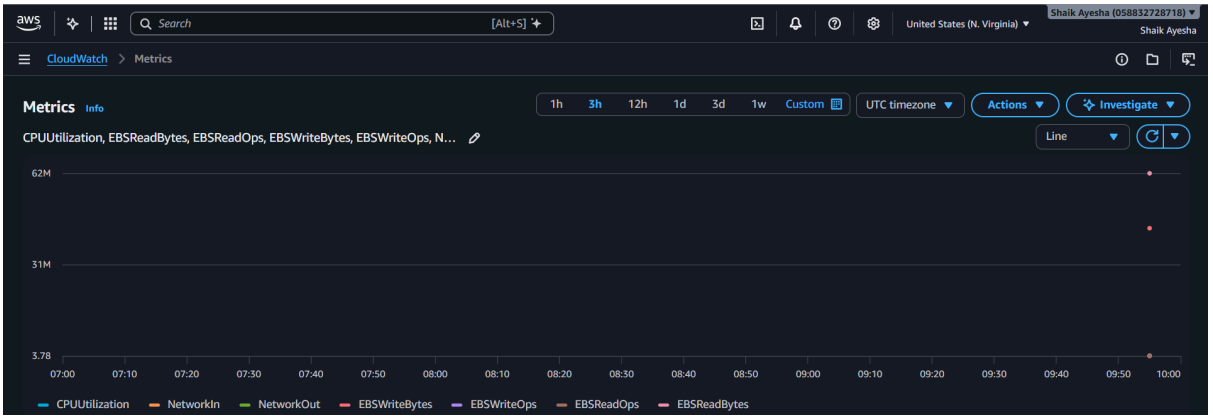
Step 5: Verify SNS Configuration

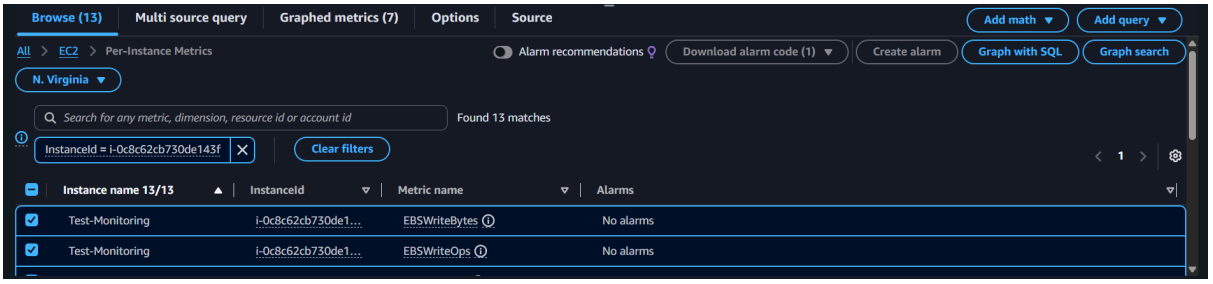
1. I returned to CloudTrail → Trails.
2. I opened My-CloudTrail.
3. I verified that SNS notification delivery was enabled.
4. I confirmed that the SNS topic was associated with the trail.

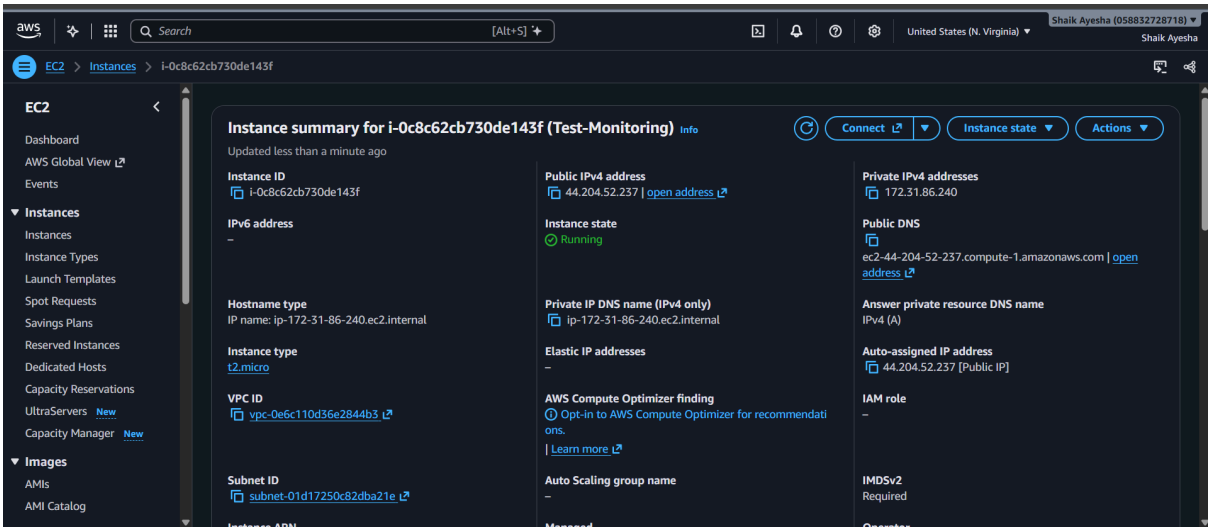
Step 6: Verify Email Notifications

1. I waited for CloudTrail to deliver a new log file.
2. I checked my email inbox.
3. I verified that I received an SNS notification.

3. Configure CloudWatch monitoring and record the CPU utilization and other metrics of EC2.







Step 1: Open CloudWatch

1. I logged in to the AWS Management Console.
2. I searched for CloudWatch.
3. I opened the CloudWatch console.
4. I selected Metrics from the left navigation panel.

Step 2: View EC2 Metrics

1. I selected All metrics.

2. I selected EC2.
3. I selected Per-Instance Metrics.
4. I selected my EC2 instance ID.
5. I viewed the available metrics.

Step 3: Monitor CPU Utilization

1. I selected the CPUUtilization metric.
2. I viewed the CPU utilization graph.
3. I selected the required time range.
4. I checked the CPU usage of my EC2 instance.

Step 4: Monitor Other EC2 Metrics

1. I selected the following metrics:
    - CPUUtilization
    - NetworkIn
    - NetworkOut
    - NetworkPacketsIn
    - NetworkPacketsOut
    - StatusCheckFailed
2. I viewed the graphs to monitor the instance performance.

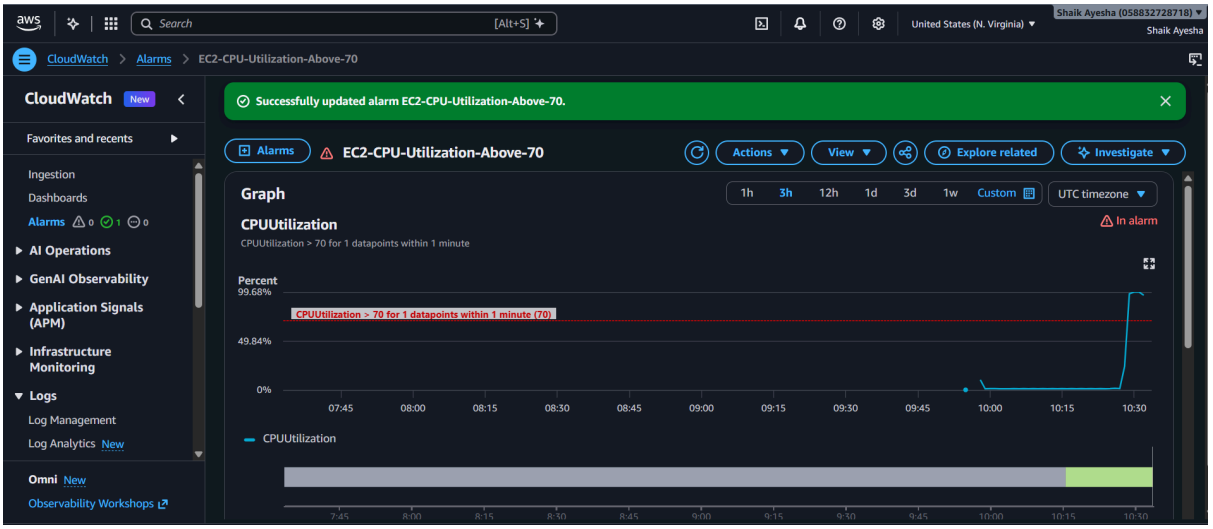
Step 5: Create a CloudWatch Dashboard

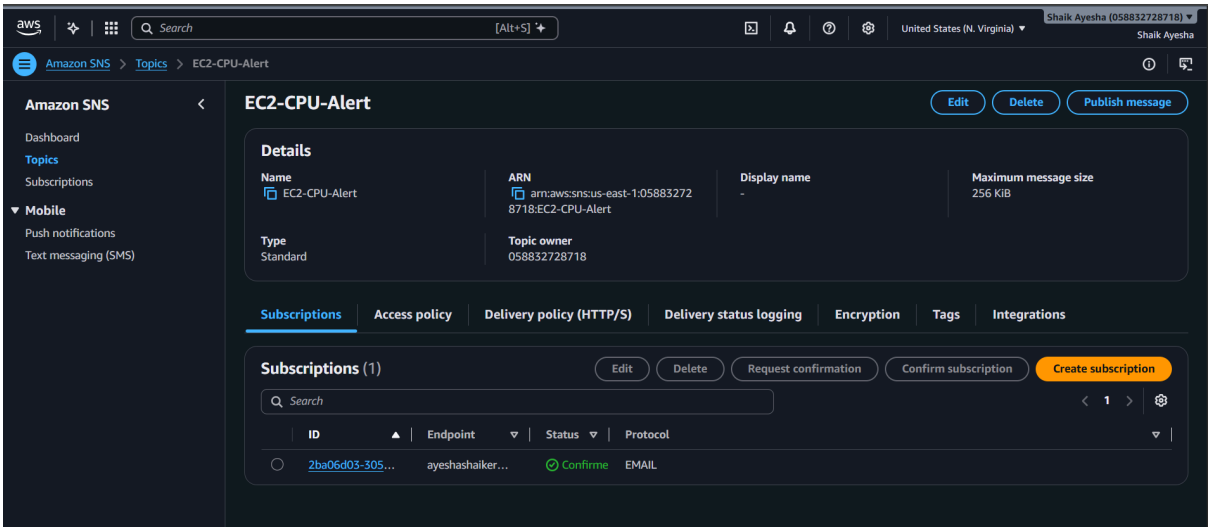
1. I navigated to Dashboards.
2. I clicked Create dashboard.
3. I entered the dashboard name EC2-Monitoring-Dashboard.
4. I selected Create dashboard.
5. I selected Line as the widget type.
6. I selected EC2 metrics.
7. I added CPUUtilization and other required metrics.
8. I clicked Create widget.
9. I saved the dashboard.

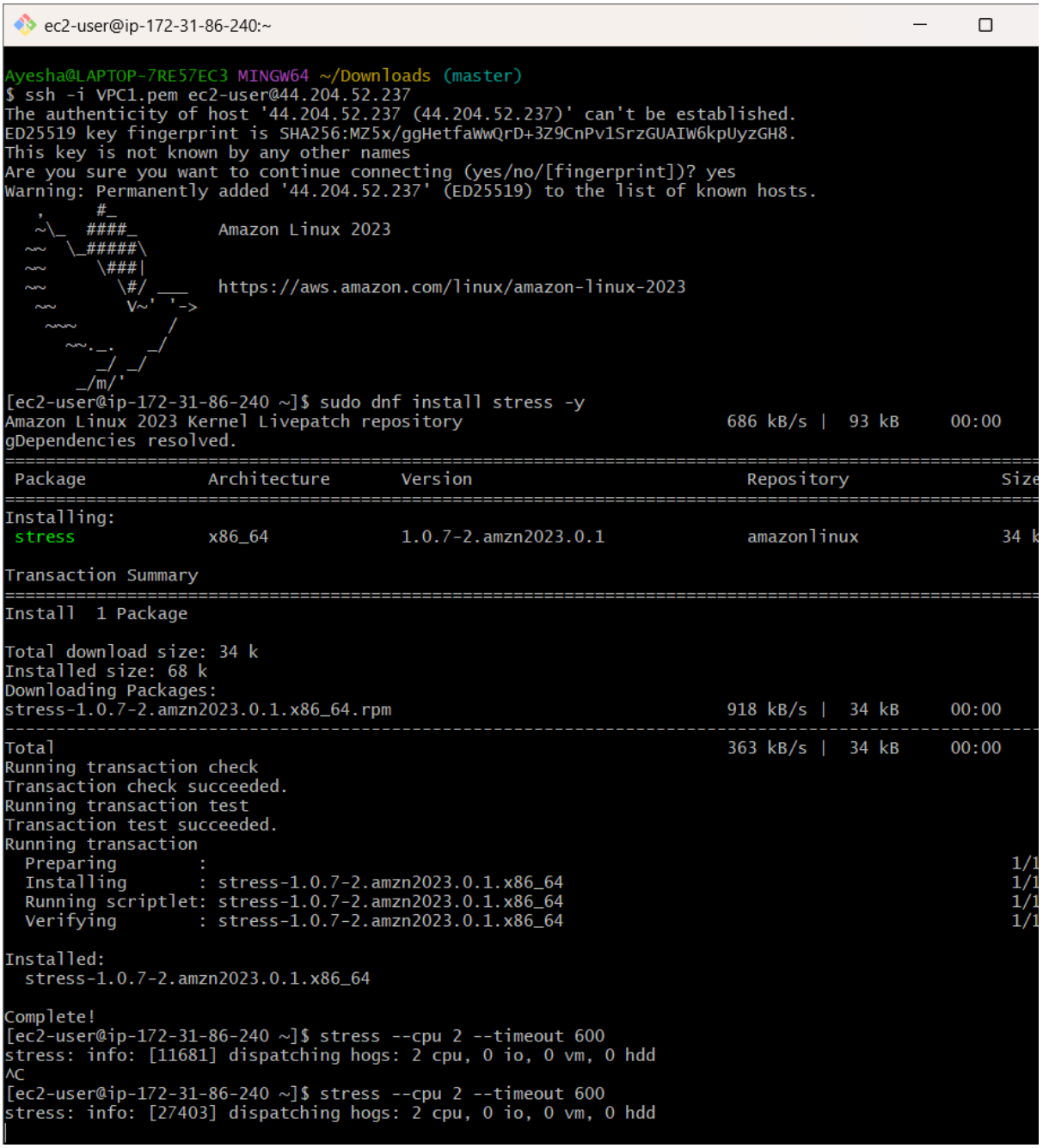
Step 6: Verify Monitoring

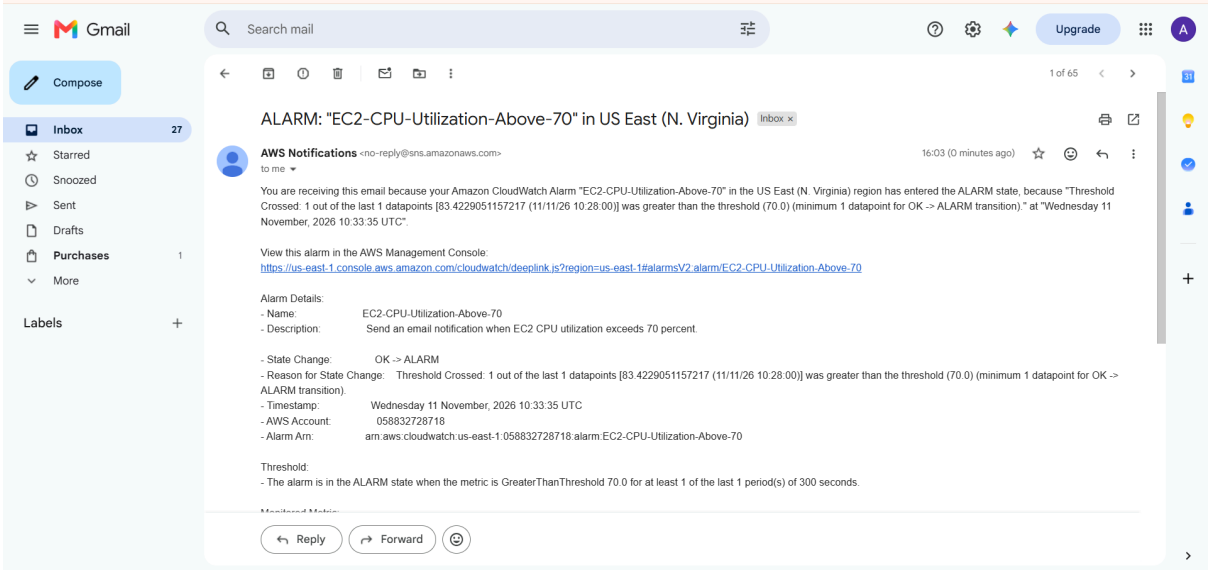
1. I opened the EC2-Monitoring-Dashboard.
2. I verified that CPU utilization and other EC2 metrics were displayed.
3. I checked the graphs to monitor EC2 performance over time.

4. Create one alarm to send an alert to email if the CPU utilization is more than 70 percent.


Search
[Alt+S]
United States (N. Virginia)
Shaik Ayesha (058832728718)
Shaik Ayesha
CloudWatch > Alarms > EC2-CPU-Utilization-Above-70
CloudWatch
New
Favorites and recents
Ingestion
Dashboards
Alarms 0 1 0
AI Operations
GenAI Observability
Application Signals (APM)
Infrastructure Monitoring
Logs
Log Management
Log Analytics New
Omni New
Observability Workshops
Successfully updated alarm EC2-CPU-Utilization-Above-70.
Alarms
EC2-CPU-Utilization-Above-70
Actions
View
Explore related
Investigate
Graph
1h 3h 12h 1d 3d 1w Custom
UTC timezone
CPUUtilization
In alarm
CPUUtilization > 70 for 1 datapoints within 1 minute
Percent
99.68%
CPUUtilization > 70 for 1 datapoints within 1 minute (70)
49.84%
0%
07:45 08:00 08:15 08:30 08:45 09:00 09:15 09:30 09:45 10:00 10:15 10:30
CPUUtilization


Search
[Alt+S]
United States (N. Virginia)
Shaik Ayesha (058832728718)
Shaik Ayesha
Amazon SNS > Topics > EC2-CPU-Alert
Amazon SNS
Dashboard
Topics
Subscriptions
Mobile
Push notifications
Text messaging (SMS)
EC2-CPU-Alert
Edit
Delete
Publish message
Details
Name
EC2-CPU-Alert
ARN
arn:aws:sns:us-east-1:058832728718:EC2-CPU-Alert
Display name
-
Maximum message size
256 KiB
Type
Standard
Topic owner
058832728718
Subscriptions
Access policy
Delivery policy (HTTP/S)
Delivery status logging
Encryption
Tags
Integrations
Subscriptions (1)
Edit
Delete
Request confirmation
Confirm subscription
Create subscription
Search
1
ID
Endpoint
Status
Protocol
2ba06d03-305...
ayeshashaiker...
Confirme
EMAIL


ec2-user@ip-172-31-86-240:~
Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/Downloads (master)
$ ssh -i VPC1.pem ec2-user@44.204.52.237
The authenticity of host '44.204.52.237 (44.204.52.237)' can't be established.
ED25519 key fingerprint is SHA256:MZ5x/ggHetfaWwQrD+3Z9CnPv1SrzGUAIW6kpUyzGH8.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '44.204.52.237' (ED25519) to the list of known hosts.
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023
[ec2-user@ip-172-31-86-240 ~]$ sudo dnf install stress -y
Amazon Linux 2023 Kernel Livepatch repository 686 kB/s | 93 kB 00:00
gDependencies resolved.
Package Architecture Version Repository Size
Installing:
stress x86_64 1.0.7-2.amzn2023.0.1 amazonlinux 34 k
Transaction Summary
Install 1 Package
Total download size: 34 k
Installed size: 68 k
Downloading Packages:
stress-1.0.7-2.amzn2023.0.1.x86_64.rpm 918 kB/s | 34 kB 00:00
Total 363 kB/s | 34 kB 00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
Preparing : 1/1
Installing : stress-1.0.7-2.amzn2023.0.1.x86_64 1/1
Running scriptlet: stress-1.0.7-2.amzn2023.0.1.x86_64 1/1
Verifying : stress-1.0.7-2.amzn2023.0.1.x86_64 1/1
Installed:
stress-1.0.7-2.amzn2023.0.1.x86_64
Complete!
[ec2-user@ip-172-31-86-240 ~]$ stress --cpu 2 --timeout 600
stress: info: [11681] dispatching hogs: 2 cpu, 0 io, 0 vm, 0 hdd
^C
[ec2-user@ip-172-31-86-240 ~]$ stress --cpu 2 --timeout 600
stress: info: [27403] dispatching hogs: 2 cpu, 0 io, 0 vm, 0 hdd



Step 1: Open CloudWatch

1. I logged in to the AWS Management Console.
2. I searched for CloudWatch.
3. I opened the CloudWatch console.
4. I selected Alarms → All alarms.
5. I clicked Create alarm.

Step 2: Select the Metric

1. I clicked Select metric.
2. I selected EC2.
3. I selected Per-Instance Metrics.
4. I selected my EC2 instance ID.
5. I selected CPUUtilization.
6. I clicked Select metric.

Step 3: Configure Alarm Conditions

1. I selected the statistic as Average.
2. I selected the period as 5 minutes.
3. Under Conditions, I selected Static.
4. I selected Greater.
5. I entered the threshold value as 70.
6. I clicked Next.

Step 4: Configure Email Notifications

1. Under Notification, I selected In alarm.
2. I selected Create new SNS topic.
3. I entered the topic name EC2-CPU-Alert.
4. I entered my email address.
5. I clicked Create topic.
6. I confirmed the subscription using the confirmation email sent to my inbox.

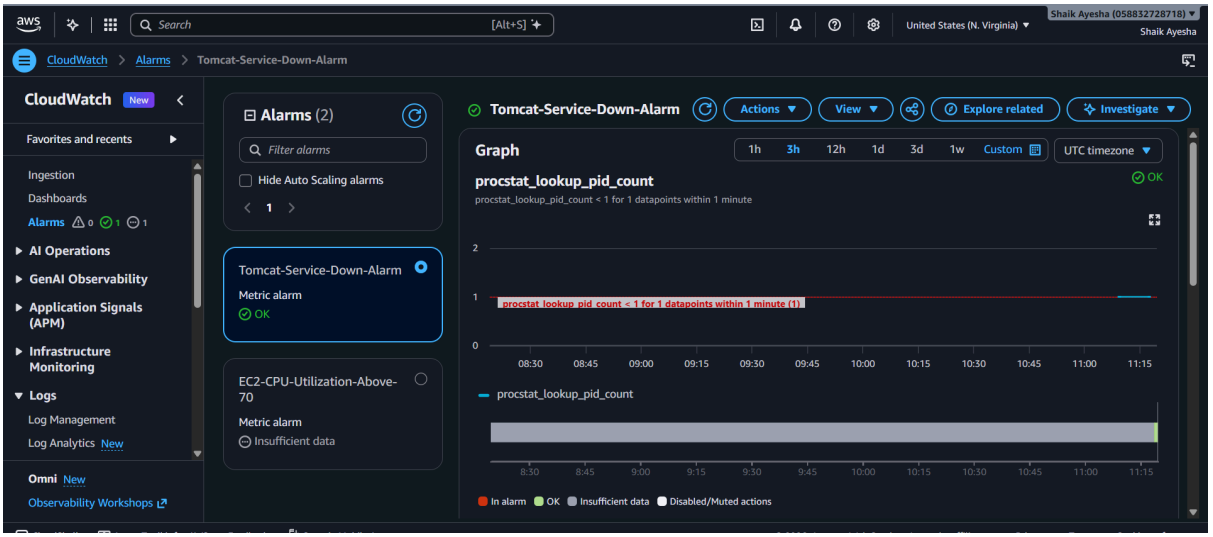
Step 5: Name the Alarm

1. I entered the alarm name EC2-CPU-Above-70.
2. I entered a description: Send an email alert when EC2 CPU utilization exceeds 70%.
3. I clicked Next.
4. I reviewed the configuration.
5. I clicked Create alarm.

Step 6: Verify the Alarm

1. I navigated to CloudWatch → Alarms → All alarms.
2. I verified that EC2-CPU-Above-70 was created.
3. I checked the alarm status.
4. I verified that the SNS email subscription was confirmed.
5. I checked my inbox for alarm notification emails.

5. Create a Dashboard and monitor the Tomcat service whether it is running or not and send the alert.



ec2-user@ip-172-31-86-149:~

```
Installed:
  apr-1.7.5-1.amzn2023.0.4.x86_64
  ecj-1:4.23-1.amzn2023.0.4.noarch
  java-17-amazon-corretto-headless-1:17.0.21.0.11-1.amzn2023.1.x86_64
  javapackages-tools-6.0.0-7.amzn2023.0.6.noarch
  tomcat-native-2.0.16-1.amzn2023.0.1.x86_64
  tomcat10-1:10.1.60-1.amzn2023.0.1.noarch
  tomcat10-el-5.0-api-1:10.1.60-1.amzn2023.0.1.noarch
  tomcat10-jsp-3.1-api-1:10.1.60-1.amzn2023.0.1.noarch
  tomcat10-lib-1:10.1.60-1.amzn2023.0.1.noarch
  tomcat10-servlet-6.0-api-1:10.1.60-1.amzn2023.0.1.noarch

Complete!
[ec2-user@ip-172-31-86-149 ~]$ rpm -qa | grep -i tomcat
tomcat10-el-5.0-api-10.1.60-1.amzn2023.0.1.noarch
tomcat10-servlet-6.0-api-10.1.60-1.amzn2023.0.1.noarch
tomcat10-jsp-3.1-api-10.1.60-1.amzn2023.0.1.noarch
tomcat10-lib-10.1.60-1.amzn2023.0.1.noarch
tomcat-native-2.0.16-1.amzn2023.0.1.x86_64
tomcat10-10.1.60-1.amzn2023.0.1.noarch
[ec2-user@ip-172-31-86-149 ~]$ sudo systemctl start tomcat
Failed to start tomcat.service: Unit tomcat.service not found.
[ec2-user@ip-172-31-86-149 ~]$ sudo systemctl start tomcat10
[ec2-user@ip-172-31-86-149 ~]$ sudo systemctl enable tomcat10
Created symlink /etc/systemd/system/multi-user.target.wants/tomcat10.service → /usr/lib/systemd/system/
tomcat10.service.
[ec2-user@ip-172-31-86-149 ~]$ sudo systemctl status tomcat10
● tomcat10.service - Apache Tomcat 10 Web Application Container
     Loaded: loaded (/usr/lib/systemd/system/tomcat10.service; enabled; preset: disabled)
     Active: active (running) since Wed 2026-11-11 10:54:51 UTC; 16s ago
   Main PID: 26920 (java)
      Tasks: 29 (limit: 1113)
     Memory: 100.3M
        CPU: 1.722s
     CGroup: /system.slice/tomcat10.service
             └─26920 /usr/lib/jvm/jre/bin/java -Djavax.sql.DataSource.Factory=org.apache.commons.dbcp.>

Nov 11 10:54:52 ip-172-31-86-149.ec2.internal server[26920]: 11-Nov-2026 10:54:52.940 INFO [main] org.>
Nov 11 10:54:52 ip-172-31-86-149.ec2.internal server[26920]: 11-Nov-2026 10:54:52.940 INFO [main] org.>
Nov 11 10:54:52 ip-172-31-86-149.ec2.internal server[26920]: 11-Nov-2026 10:54:52.948 INFO [main] org.>
Nov 11 10:54:52 ip-172-31-86-149.ec2.internal server[26920]: 11-Nov-2026 10:54:52.960 INFO [main] org.>
Nov 11 10:54:53 ip-172-31-86-149.ec2.internal server[26920]: 11-Nov-2026 10:54:53.691 INFO [main] org.>
Nov 11 10:54:53 ip-172-31-86-149.ec2.internal server[26920]: 11-Nov-2026 10:54:53.761 INFO [main] org.>
Nov 11 10:54:53 ip-172-31-86-149.ec2.internal server[26920]: 11-Nov-2026 10:54:53.864 INFO [main] org.>
Nov 11 10:54:53 ip-172-31-86-149.ec2.internal server[26920]: 11-Nov-2026 10:54:53.867 INFO [main] org.>
Nov 11 10:54:53 ip-172-31-86-149.ec2.internal server[26920]: 11-Nov-2026 10:54:53.895 INFO [main] org.>
Nov 11 10:54:53 ip-172-31-86-149.ec2.internal server[26920]: 11-Nov-2026 10:54:53.921 INFO [main] org.>

[ec2-user@ip-172-31-86-149 ~]$ sudo ss -lntp | grep 8080
LISTEN 0      100                *:8080           *:*    users:(("java",pid=26920,fd=48))
[ec2-user@ip-172-31-86-149 ~]$ ls -l /var/lib/tomcat/webapps/
ls: cannot access '/var/lib/tomcat/webapps/': No such file or directory
[ec2-user@ip-172-31-86-149 ~]$ ls -l /var/lib/tomcat10/webapps/
total 0
[ec2-user@ip-172-31-86-149 ~]$ sudo systemctl status tomcat10
● tomcat10.service - Apache Tomcat 10 Web Application Container
     Loaded: loaded (/usr/lib/systemd/system/tomcat10.service; enabled; preset: disabled)
```

```
ec2-user@ip-172-31-86-149:~
-a fetch-config \
-m ec2 \
-c file:/opt/aws/amazon-cloudwatch-agent/bin/config.json \
-s
****** processing amazon-cloudwatch-agent ******
Starting config-downloader, this will map back to a call to amazon-cloudwatch-agent
Executing /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent with arguments: [config-download
er -download-source file:/opt/aws/amazon-cloudwatch-agent/bin/config.json -output-dir /opt/aws/amazon-c
loudwatch-agent/etc/amazon-cloudwatch-agent.d -config /opt/aws/amazon-cloudwatch-agent/etc/common-confi
g.toml -multi-config default -mode ec2]I! Trying to detect region from ec2
D! [EC2] Found active network interface
I! imds retry client will retry 1 times
Start configuration validation...
2026/11/11 11:08:59 Reading json config file path: /opt/aws/amazon-cloudwatch-agent/etc/amazon-cloudwat
ch-agent.d/file_config.json.tmp ...
2026/11/11 11:08:59 I! Valid Json input schema.
2026/11/11 11:08:59 D! ec2tagger processor required because append_dimensions is set
2026/11/11 11:08:59 Configuration validation first phase succeeded
Starting config-translator, this will map back to a call to amazon-cloudwatch-agent
Executing /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent with arguments: [config-translat
or -config /opt/aws/amazon-cloudwatch-agent/etc/common-config.toml -multi-config default -input /opt/aw
s/amazon-cloudwatch-agent/etc/amazon-cloudwatch-agent.json -input-dir /opt/aws/amazon-cloudwatch-agent/
etc/amazon-cloudwatch-agent.d -output /opt/aws/amazon-cloudwatch-agent/etc/amazon-cloudwatch-agent.toml
 -mode ec2]I! Trying to detect region from ec2
D! [EC2] Found active network interface
I! imds retry client will retry 1 times
/opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent -schematest -config /opt/aws/amazon-cloudw
atch-agent/etc/amazon-cloudwatch-agent.toml
Configuration validation second phase succeeded
Configuration validation succeeded
amazon-cloudwatch-agent has already been stopped
Created symlink /etc/systemd/system/multi-user.target.wants/amazon-cloudwatch-agent.service → /etc/syst
emd/system/amazon-cloudwatch-agent.service.
[ec2-user@ip-172-31-86-149 ~]$ sudo systemctl status amazon-cloudwatch-agent
● amazon-cloudwatch-agent.service - Amazon CloudWatch Agent
     Loaded: loaded (/etc/systemd/system/amazon-cloudwatch-agent.service; enabled; preset: disabled)
     Active: active (running) since Wed 2026-11-11 11:08:59 UTC; 5s ago
   Main PID: 27768 (amazon-cloudwat)
      Tasks: 5 (limit: 1113)
     Memory: 47.3M
        CPU: 190ms
     CGroup: /system.slice/amazon-cloudwatch-agent.service
             └─27768 /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent -config /opt/aws/ama>

Nov 11 11:08:59 ip-172-31-86-149.ec2.internal start-amazon-cloudwatch-agent[27776]: D! [EC2] Found act>
Nov 11 11:08:59 ip-172-31-86-149.ec2.internal start-amazon-cloudwatch-agent[27776]: I! imds retry clie>
Nov 11 11:08:59 ip-172-31-86-149.ec2.internal start-amazon-cloudwatch-agent[27776]: 2026/11/11 11:08:5>
Nov 11 11:08:59 ip-172-31-86-149.ec2.internal start-amazon-cloudwatch-agent[27776]: /opt/aws/amazon-cl>
Nov 11 11:08:59 ip-172-31-86-149.ec2.internal start-amazon-cloudwatch-agent[27776]: 2026/11/11 11:08:5>
Nov 11 11:08:59 ip-172-31-86-149.ec2.internal start-amazon-cloudwatch-agent[27776]: 2026/11/11 11:08:5>
Nov 11 11:08:59 ip-172-31-86-149.ec2.internal start-amazon-cloudwatch-agent[27776]: I! Trying to detec>
Nov 11 11:08:59 ip-172-31-86-149.ec2.internal start-amazon-cloudwatch-agent[27776]: 2026/11/11 11:08:5>
Nov 11 11:08:59 ip-172-31-86-149.ec2.internal start-amazon-cloudwatch-agent[27776]: 2026/11/11 11:08:5>
Nov 11 11:08:59 ip-172-31-86-149.ec2.internal start-amazon-cloudwatch-agent[27768]: I! Detecting run_a>
[ec2-user@ip-172-31-86-149 ~]$ sudo systemctl stop tomcat
Failed to stop tomcat.service: Unit tomcat.service not loaded.
[ec2-user@ip-172-31-86-149 ~]$ sudo systemctl stop tomcat10
[ec2-user@ip-172-31-86-149 ~]$
```

2 of 66

ALARM: "Tomcat-Service-Down-Alarm" in US East (N. Virginia)

Inbox ×

AWS Notifications <no-reply@sns.amazonaws.com> 16:50 (0 minutes ago)

to me

You are receiving this email because your Amazon CloudWatch Alarm "Tomcat-Service-Down-Alarm" in the US East (N. Virginia) region has entered the ALARM state, because "Threshold Crossed: 1 out of the last 1 datapoints [0.0 (11/11/26 11:19:00)] was less than the threshold (1.0) (minimum 1 datapoint for OK -> ALARM transition)." at "Wednesday 11 November, 2026 11:20:36 UTC".

View this alarm in the AWS Management Console:
https://us-east-1.console.aws.amazon.com/cloudwatch/deeplink.js?region=us-east-1#alarmsV2:alarm/Tomcat-Service-Down-Alarm

Alarm Details:
- Name: Tomcat-Service-Down-Alarm
- Description: Send email alert when Tomcat service stops running.
- State Change: OK -> ALARM
- Reason for State Change: Threshold Crossed: 1 out of the last 1 datapoints [0.0 (11/11/26 11:19:00)] was less than the threshold (1.0) (minimum 1 datapoint for OK -> ALARM transition).
- Timestamp: Wednesday 11 November, 2026 11:20:36 UTC
- AWS Account: 058832728718
- Alarm Arn: arn:aws:cloudwatch:us-east-1:058832728718:alarm:Tomcat-Service-Down-Alarm

Threshold:
- The alarm is in the ALARM state when the metric is LessThanThreshold 1.0 for at least 1 of the last 1 period(s) of 60 seconds.

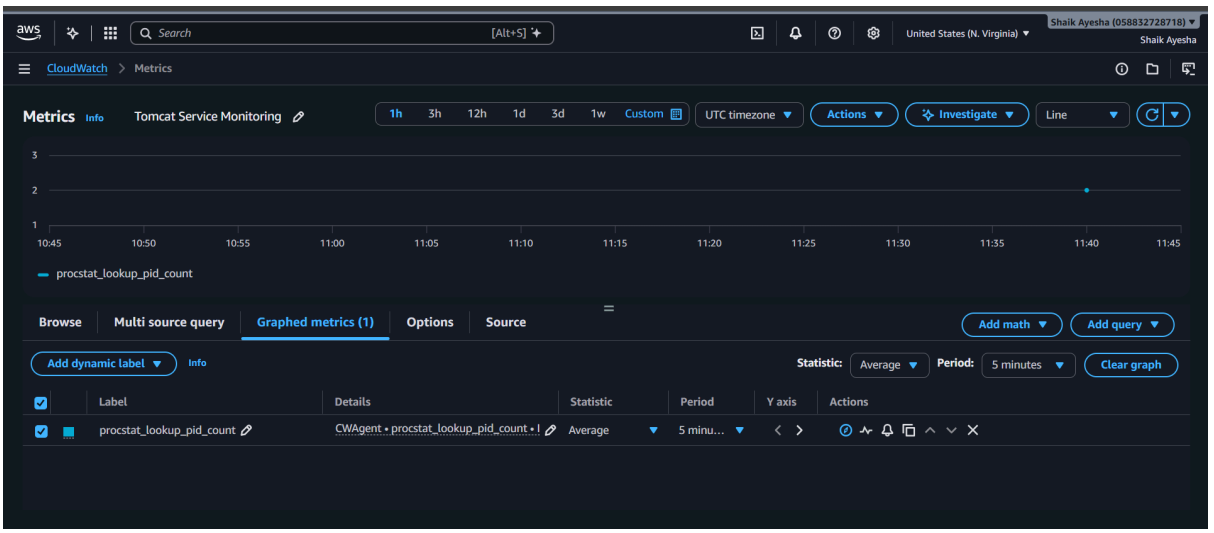
Monitored Metric:
- MetricNamespace: CWAgent

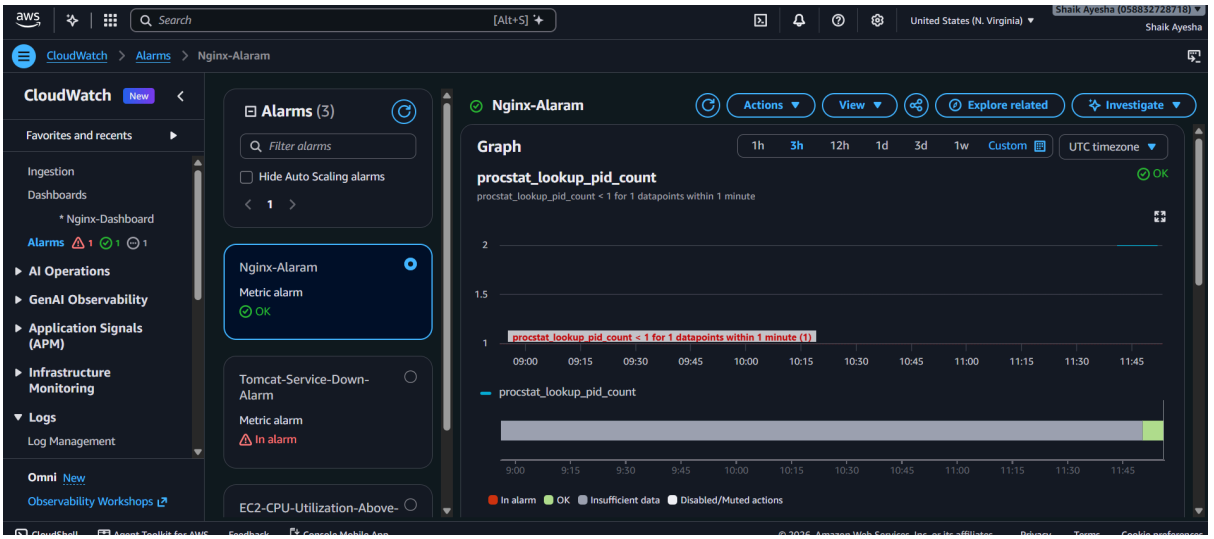
Reply Forward

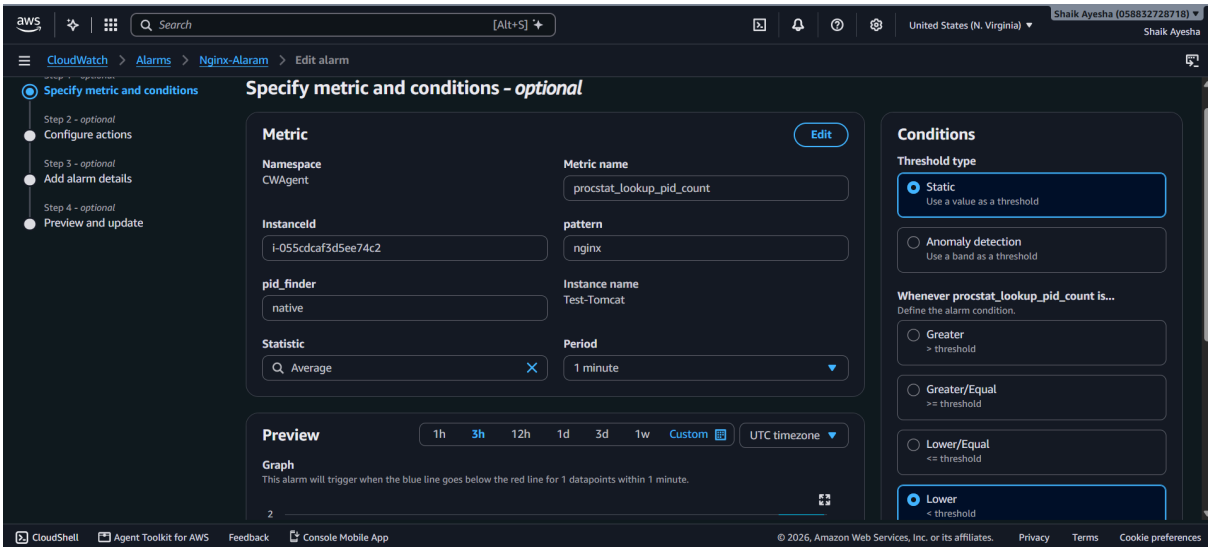


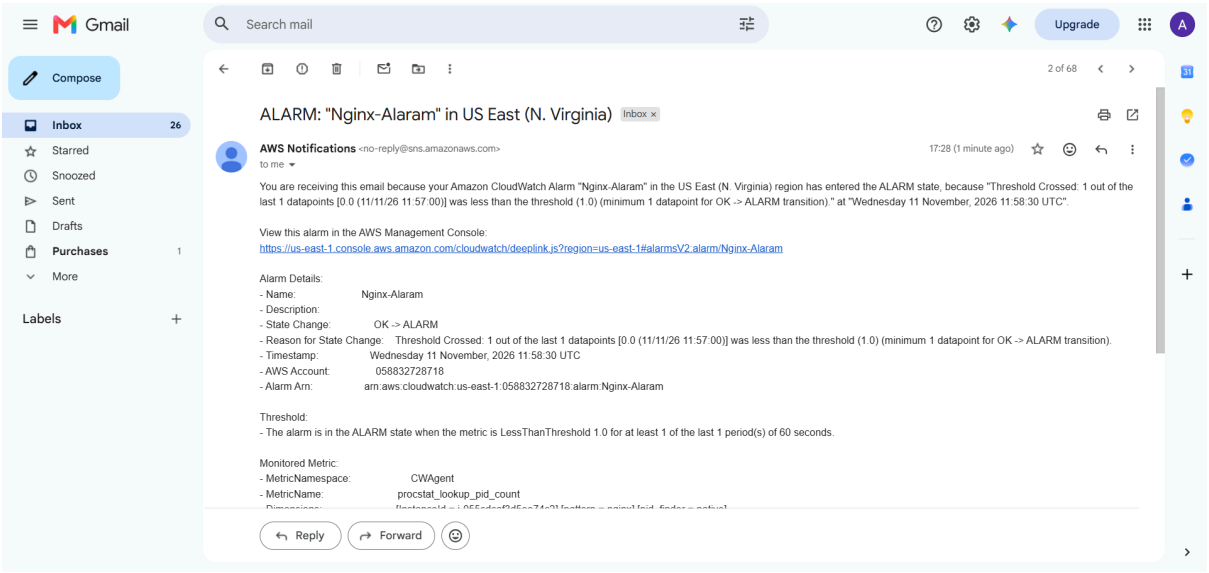
1. I installed Tomcat on my EC2 instance.
2. I started the Tomcat service and verified that it was running.
3. I created an IAM role with CloudWatchAgentServerPolicy.
4. I attached the IAM role to my EC2 instance.
5. I installed the Amazon CloudWatch Agent on my EC2 instance.
6. I created the CloudWatch Agent configuration file to monitor the Tomcat process.
7. I configured the metrics collection interval to 60 seconds.
8. I started the CloudWatch Agent and verified its status.
9. I opened CloudWatch and verified that the Tomcat process metrics were available.
10. I created an SNS topic named Tomcat-Service-Alerts.
11. I subscribed my email address to the SNS topic.
12. I confirmed the SNS subscription through the email received.
13. I created a CloudWatch alarm named Tomcat-Service-Down-Alert.
14. I configured the alarm to trigger when the Tomcat process count falls below 1.
15. I configured the alarm to send email notifications through SNS.
16. I created a CloudWatch Dashboard named Tomcat-Monitoring-Dashboard.
17. I added Tomcat process count, CPU usage, memory usage, and alarm status widgets to the dashboard.
18. I stopped the Tomcat service to test the alarm.
19. I verified that the CloudWatch alarm changed to ALARM state and received an email notification.
20. I restarted the Tomcat service and verified that the alarm returned to OK state.

6. Create a Dashboard and monitor the Nginx service to send the alert if Nginx is not running.


Search
[Alt+S]
United States (N. Virginia)
Shaik Ayesha
CloudWatch > Metrics
Metrics Info
Tomcat Service Monitoring
1h 3h 12h 1d 3d 1w Custom
UTC timezone
Actions
Investigate
Line
procstat_lookup_pid_count
Browse
Multi source query
Graphed metrics (1)
Options
Source
Add math
Add query
Add dynamic label
Info
Statistic: Average
Period: 5 minutes
Clear graph
Label
Details
Statistic
Period
Y axis
Actions
procstat_lookup_pid_count
CWAgent • procstat_lookup_pid_count • I
Average
5 minu...


Search
[Alt+S]
United States (N. Virginia)
Shaik Ayesha
CloudWatch > Alarms > Nginx-Alaram
CloudWatch New
Favorites and recents
Ingestion
Dashboards
* Nginx-Dashboard
Alarms
AI Operations
GenAI Observability
Application Signals (APM)
Infrastructure Monitoring
Logs
Log Management
Omni New
Observability Workshops
Alarms (3)
Filter alarms
Hide Auto Scaling alarms
Nginx-Alaram
Metric alarm
OK
Tomcat-Service-Down-Alarm
Metric alarm
In alarm
EC2-CPU-Utilization-Above-
Nginx-Alaram
Actions
View
Explore related
Investigate
Graph
1h 3h 12h 1d 3d 1w Custom
UTC timezone
procstat_lookup_pid_count
procstat_lookup_pid_count < 1 for 1 datapoints within 1 minute
OK
procstat_lookup_pid_count < 1 for 1 datapoints within 1 minute (1)
procstat_lookup_pid_count
In alarm
OK
Insufficient data
Disabled/Muted actions


Search
[Alt+S]
United States (N. Virginia)
Shaik Ayesha
CloudWatch > Alarms > Nginx-Alaram > Edit alarm
Specify metric and conditions
Step 2 - optional
Configure actions
Step 3 - optional
Add alarm details
Step 4 - optional
Preview and update
Specify metric and conditions - optional
Metric
Edit
Namespace
CWAgent
Metric name
procstat_lookup_pid_count
InstanceId
i-055cdcaf3d5ee74c2
pattern
nginx
pid_finder
native
Instance name
Test-Tomcat
Statistic
Average
Period
1 minute
Preview
1h 3h 12h 1d 3d 1w Custom
UTC timezone
Graph
This alarm will trigger when the blue line goes below the red line for 1 datapoints within 1 minute.
Conditions
Threshold type
Static
Use a value as a threshold
Anomaly detection
Use a band as a threshold
Whenever procstat_lookup_pid_count is...
Define the alarm condition.
Greater
> threshold
Greater/Equal
>= threshold
Lower/Equal
<= threshold
Lower
< threshold
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
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